DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS / IT

Calculus and Optimization

Maxima and Minima

Lecture No. 03



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Recap of previous lecture



Topic

Maxima-Minima (Part 2)

(Discussion of THEORY)

Topics to be Covered



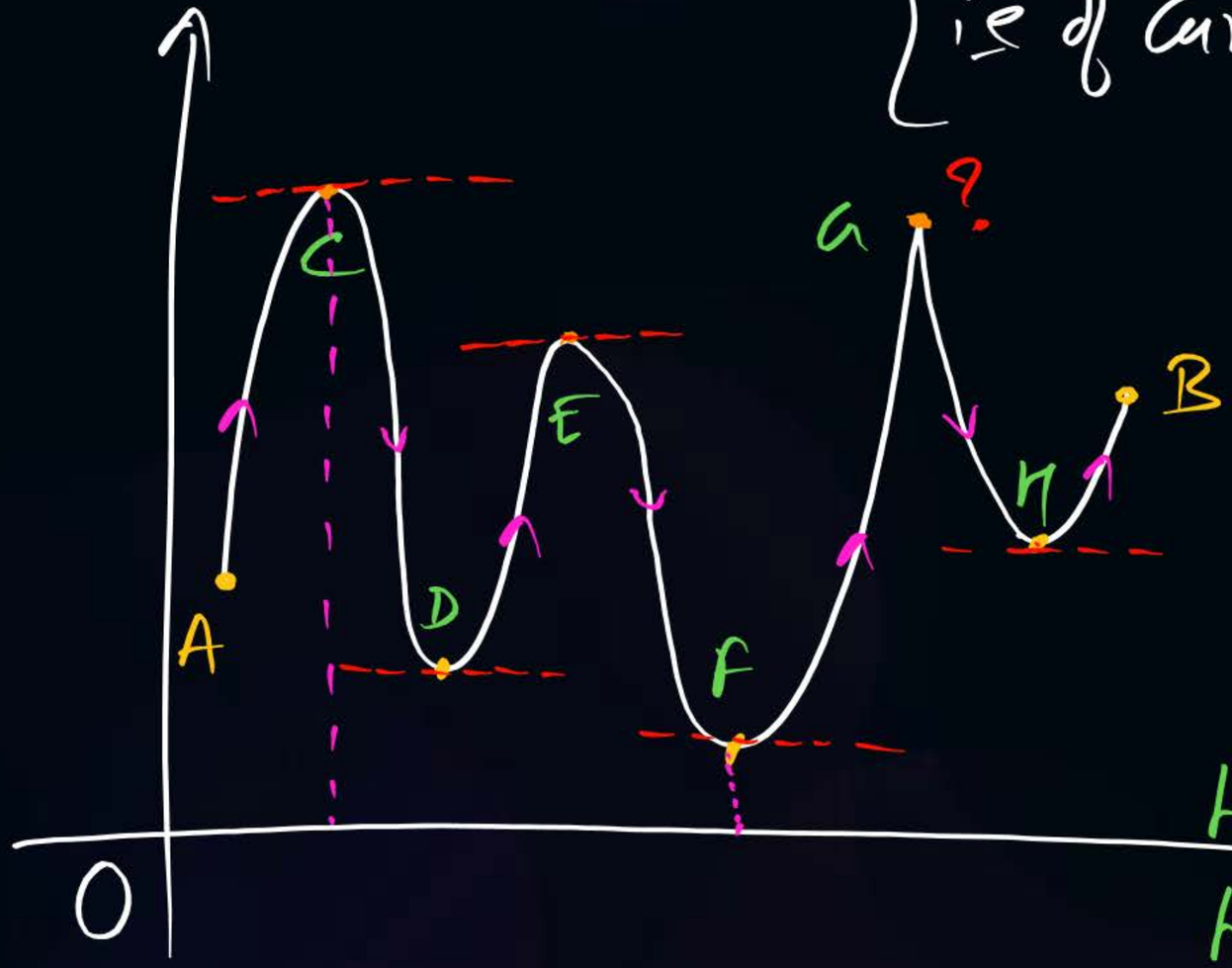
Topic MAXIMA-MINIMA (Part 3)

→ Practice Questions

→ Max-Min of Surface (i.e $Z=f(x,y)$)

MAXIMA-MINIMA of Func^n of Single Variable

[i.e. of Curve $y=f(x)$.] → 12th Class.



Point $\simeq x$

Value $\simeq y$

Local Max Points → C, E, G, B

Local Max Values → f(C), f(E), f(G), f(B)

Absolute Maxima / Global → f(C)

Local min Points → A, D, F, H.

Local min Values → f(A), f(D), f(F), f(H).

Absolute Minima / Global → f(F)

① Necessary Condition for Maxima-Minima → $f'(x) = 0$ or Not Defined (T)

② Stationary Points: eg C, D, E, F, H

those points where tangent is Horizontal are called Stationary Points.

ie at Stationary Points, $f'(x) = 0$

③ Turning Points / critical Points → eg C, D, E, F, H, G

those points where tangent is either Horizontal or Not defined are Called T. Points. ie at T. Points, $f'(x) = 0$ or undefined (T) True

→ All the Stationary Points are Turning Points But Converse is not Necessarily (T)

→ By solving eqn (1) we can find Turning Points.

T-Points.

④ Extreme Points / Optimal Points → eg A, C, D, E, F, G, H, B

those points where we get either Maxima or Minima are called E-Points

→ Corner Points are also Extreme Points (True)

But take Care; Corner Points are not Critical Points.

⑤ Points of Inflexion / Saddle points →

those points where Curve Changes it's Concavity are called Saddle Points

⑥ Maxima & Minima occurs alternately (True)

⑦ Sufficient Cond^{n} for Maxima-Minima → f(x) must satisfy either

1st Derivative test or 2nd Derivative test.

(8) 1st Derivative test :- Let $x = c$ is the Turning Point.

(it is obtained by solving equ^n (1) i.e $f'(c) = 0$ or undefined.

Now we will check the sign of $f'(c)$ as follows ; M. Imp.



(Maxima)



(minima)

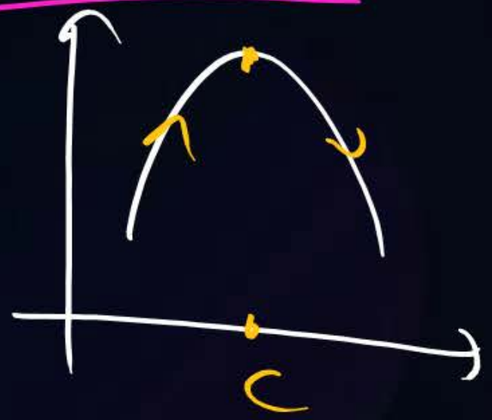


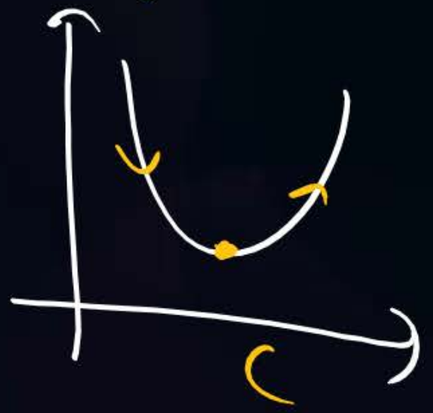
(NNNM)



(NMNM)

Explanation :-









(9) 2nd Derivative test → Let $x=c$ is Critical Point then

$f'(c)=0$

If $f^n(c)=0$ & $f^{n+1}(c)\neq 0$;
n = even then
$x=c$ is Point of Inflexion

$f''(c)>0$

($x=c$ is Point of Minima)

$\Downarrow$

$f(x)$ is minimum at $x=c$

$f''(c)=0$

$\downarrow$

$f'''(c)\neq 0$

($x=c$ is Saddle point)

$f''(c)<0$

($x=c$ is Point of Maxima)

$\Downarrow$

$f(x)$ is Maximum at $x=c$

(10) If we want to check the Nature of Corner points, then we have (No) Shortcut Method 😕

it's Nature can be Calculated by using following theorem only

"Maxima & Minima occurs Alternately"

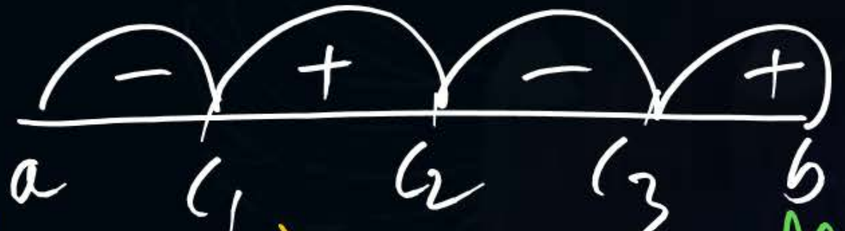
Eg. Let $f(x)$ is defined in $[a,b]$ & it's T. Points are $x=c_1,c_2,c_3$

Case I:



Sign of $f'(x)$

Case II:



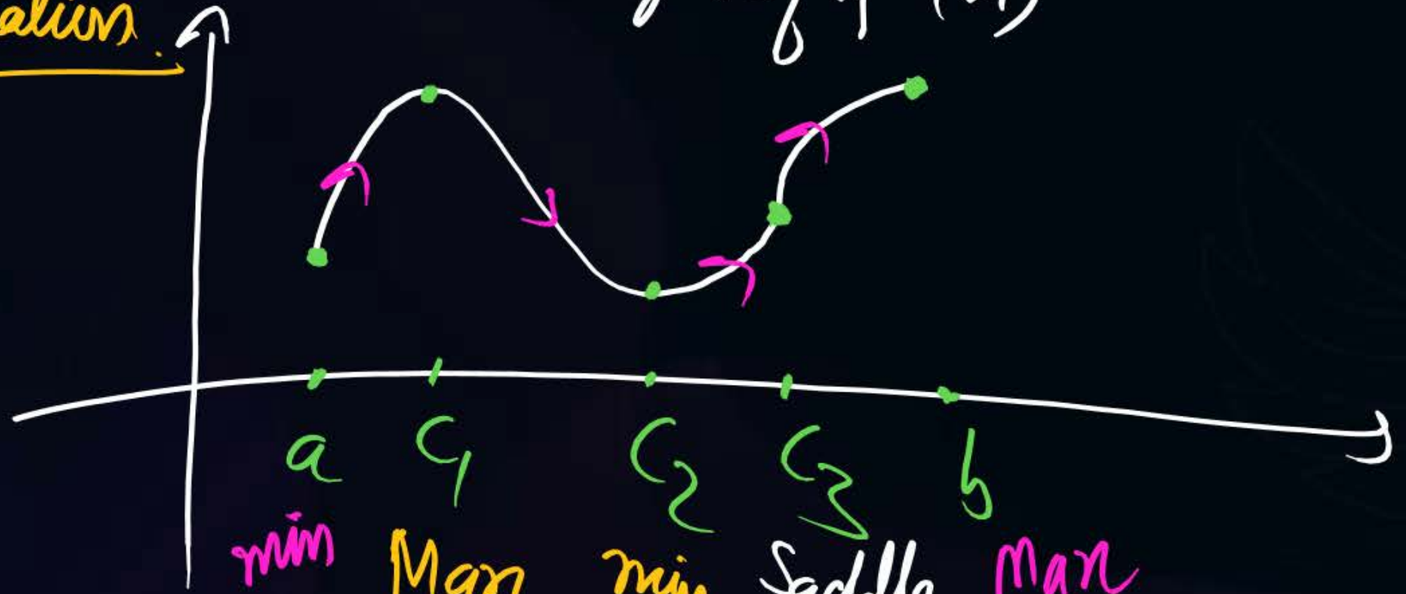
Sign of $f'(x)$

Case II :



Sign of $f'(x)$

Explanation :



Questions Based on type I :- (Common sense Based Questions)

Q1:- Find Maxima & Minima of ① $y = x^2 + 5$ ② $y = \sin 3x + 4$, ③ $y = |x+3|$

Sol: ① $f(x) = x^2 + 5$: $D_f = (-\infty, \infty)$



Point of Minima is $x = 0$

Minimum Value is $y = 5$

Maximum Value is DNE

& Point of Maxima : Question Wrong

Sol ② $f(x) = \sin 3x + 4$

W.K.T.hat,

$$-1 \le \sin 3x \le 1$$

$$-1+4 \le \sin 3x + 4 \le 1+4$$

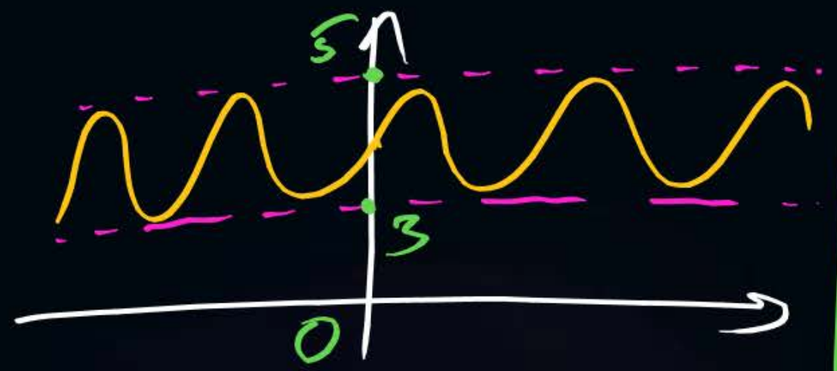
$$3 \le \sin 3x + 4 \le 5$$

$$3 \le f(x) \le 5$$

ie Min Value is 3

Max .. is 5.

Graph: $y = \sin 3x + 4$



(3) $y = f(x) = |x+3|$

W.K that $|x+3| \geq 0$

So $f(x) \geq 0$

Min Value = 0 & Max Value = DNE

(M-II) $y = |x+3|$



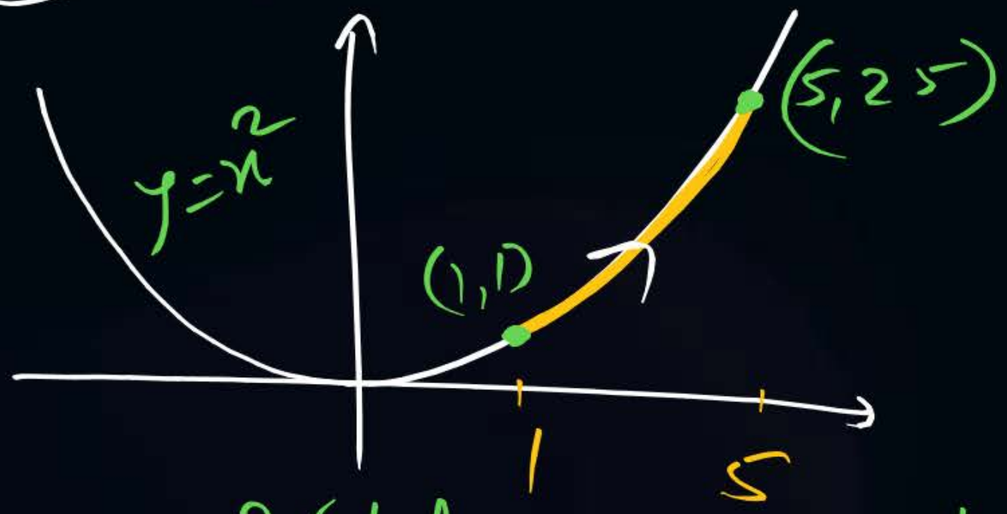
Point of minima is $x = -3$

Min Value is $y = 0$

Max Value = ND

GATE Max Value of $f(x)=x^2$ in $[1,5]$ will be ?

M-I $y=x^2$; $[1,5]$



Point of minima is $x=1$

∴ Maxima is $x=5$

Min Value is $f(1)=1$

Max Value $f(5)=25$

M-II $f(x)=x^2$, $[1,5]$

$f'(x)=2x$

C.P. are, $f'(x)=0 \Rightarrow x=0$

$\because x=0 \notin$ given Domain of f.

So $f(x)$ has No Turning Point.



i.e. $f(x)$ is S.Inc in $[1,5]$

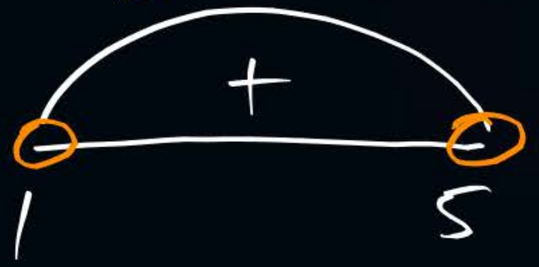
So Min Value occurs at $x=1$

& Max Value will occur at $x=5$

Q: Find Min & Max Values of $f(x) = x^2$ in $(1,5)$?

Sol: $f(x) = x^2$

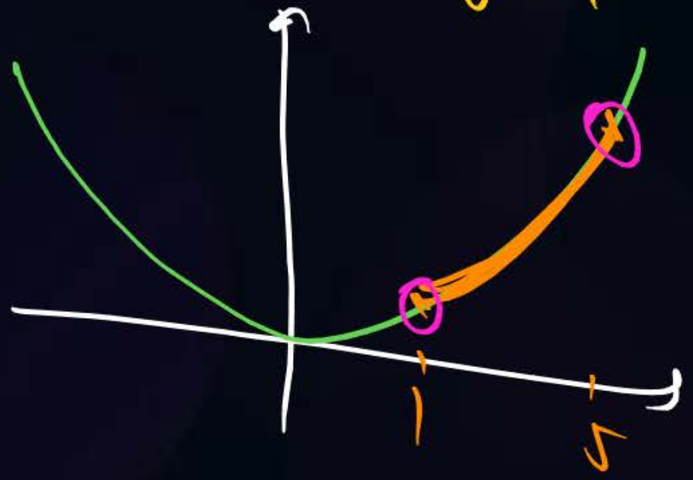
$f'(x) = 2x$



Min Value = N.D

Max Value = N.D

∵ In this Question, we have No Turning points as well as No Corner Points that's why Not able to find Min & Max Values.



$1 < f(x) < 25$

But Min $f(x) = ?$

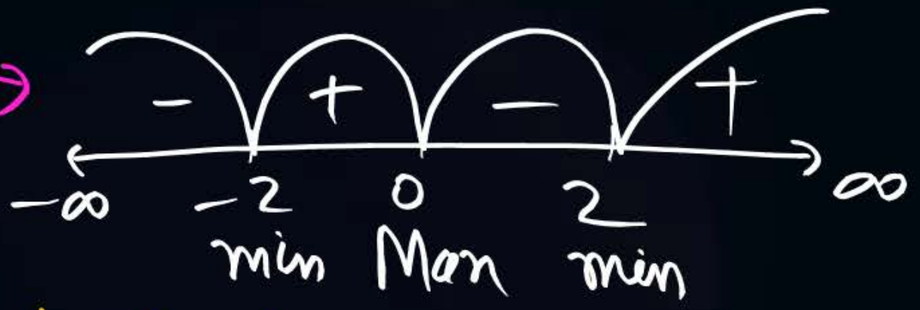
& Max $f(x) = ?$

Type II (Basically Bascalan 1st Derivative test) ⟶

Q $y=(x^2-4)^2, (-\infty, \infty)$

Sol: $f'(x)=4x(x+2)(x-2)$

T-Points are $x=-2, 0, 2$



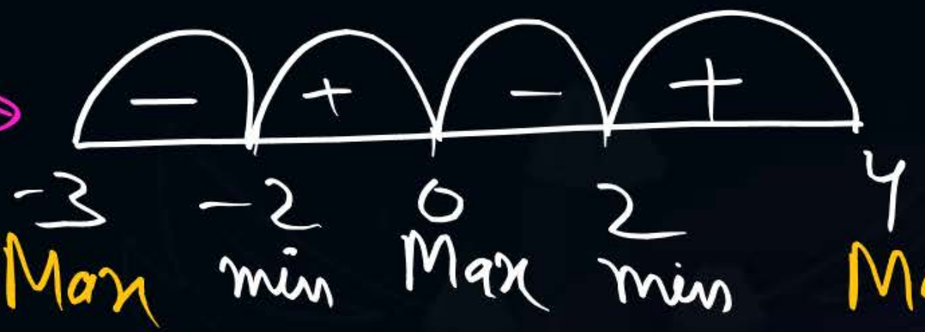
① Number of extremas are three

Note: $f(x)=(x^2-4)^2 \begin{cases} f(0)=16 \\ f(-3)=25 \ \& \ f(4)=144 \end{cases}$

Q: $y=(x^2-4)^2, [-3, 4]$

$f'(x)=4x(x+2)(x-2)$

T-Points are $x=-2, 0, 2$



① Number of extremas are FIVE

② Points of L. Maxima is; $x=0$

③ Local Max Value $= f(0) = 16$

④ Absolute Maxima is $=$ DNE

⑤ Local Min Points $x = \pm 2$

⑥ Local Min Values $\begin{cases} f(-2)=0 \\ f(2)=0 \end{cases}$

⑦ Absolute minima $= 0$

⑧ Points of Inflexion are $x = \frac{\pm 2}{\sqrt{3}}$

② Points of Local Maxima $x = -3, 0, 4$

③ Local Max Values $\begin{cases} f(-3)=25 \\ f(0)=16 \\ f(4)=144 \end{cases}$

④ Absolute Maxima is $= 144$

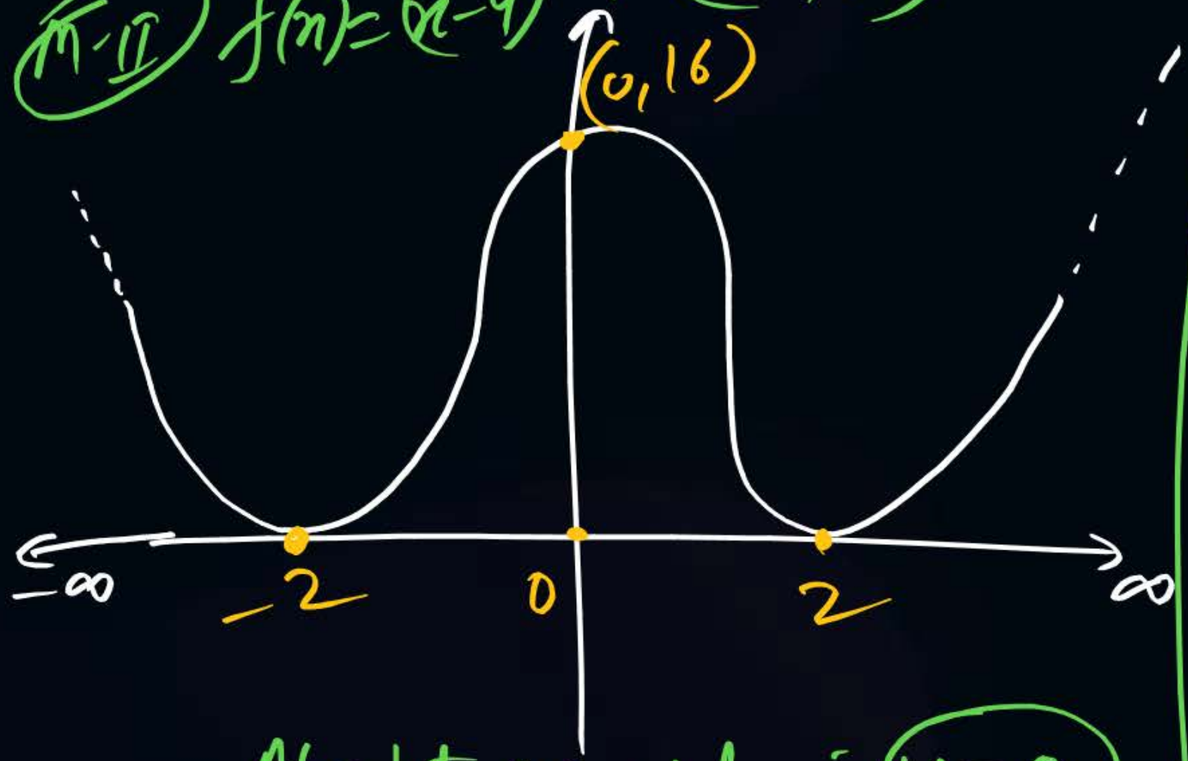
⑤ Local Min Points $x = \pm 2$

⑥ Local Min Values $\begin{cases} f(-2)=0 \\ f(2)=0 \end{cases}$

⑦ Absolute Minima $= 0$

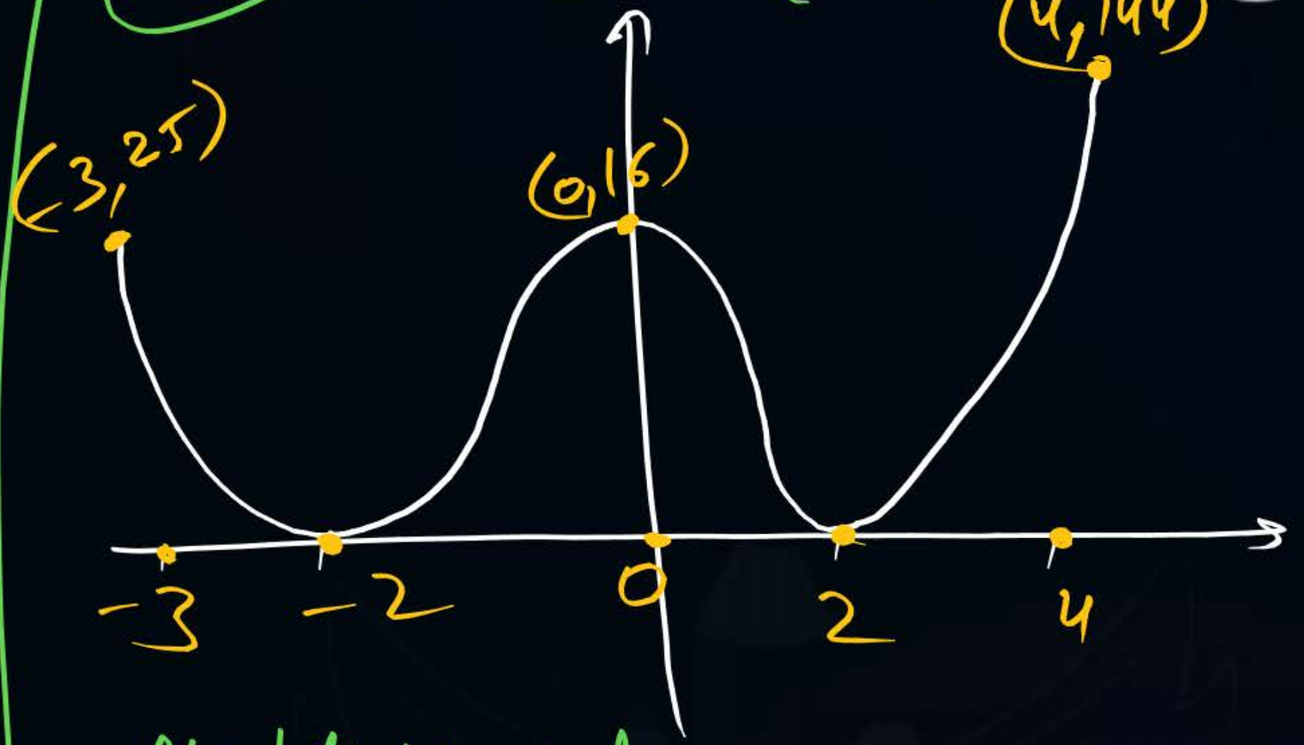
⑧ Points of Inflexion are $x = \pm \frac{2}{\sqrt{3}}$

M-II $f(x)=(x^2-4)^2$, $(-\infty,\infty)$



Absolute Min. Value is $y=0$
& it occurs at $x=+2$ or -2
Absolute Max Value = DNE

M-II $f(x)=(x^2-4)^2$, $[-3,4]$



Absolute Max Value = 144

Q: $f(x)=x^3-9x^2+24x+5$, defined in $[1,6]$ then Max Value = ?

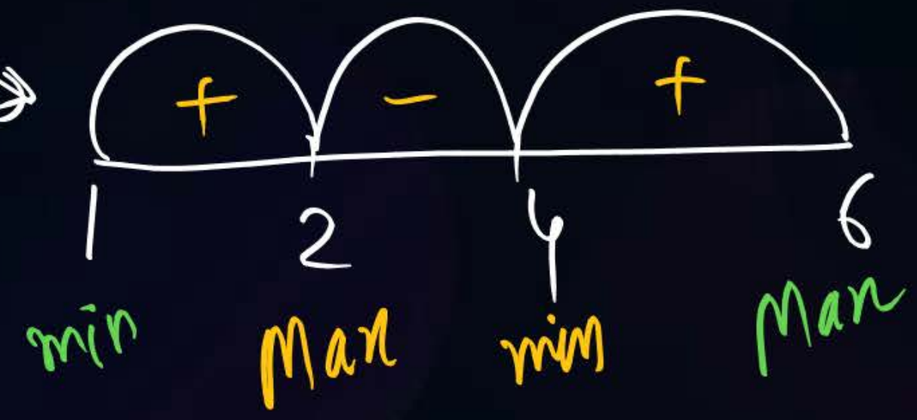
Sol: $f'(x)=3x^2-18x+24$

$=3(x^2-6x+8)$

$=3(x^2-4x-2x+8)$

$f'(x)=3(x-4)(x-2)$

Critical Points are, $x=2$ & 4



Local Max Points $\begin{cases} x=2 \\ x=6 \end{cases}$

Local Max Values $\begin{cases} f(2)=25 \\ f(6)=41 \end{cases}$

So Absolute Max Value $= 41$ Ans

Note: Local Min Points $\begin{cases} x=1 \\ x=4 \end{cases}$

Local Min Values $\begin{cases} f(1)=21 \\ f(4)=21 \end{cases}$

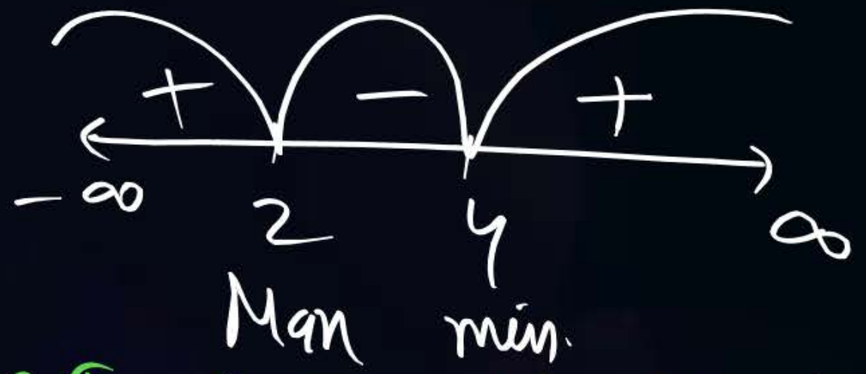
Absolute Min Value $= 21$

Q: If $f(x)=x^3-9x^2+24x+5$ then Find Min & Max Values of $f(x)$?

Sol: $f(x)=x^3-9x^2+24x+5$, $(-\infty,\infty)$

$$\boxed{f'(x)=3(x-4)(x-2)}$$

T. Points are $x=2$ & 4.



$f(x)$ S.Inc in $(-\infty,2)\cup(4,\infty)$

$f(x)$ S.Dec in $(2,4)$

Local Max Point is $x=2$

" " Value is $y=f(2)=25$

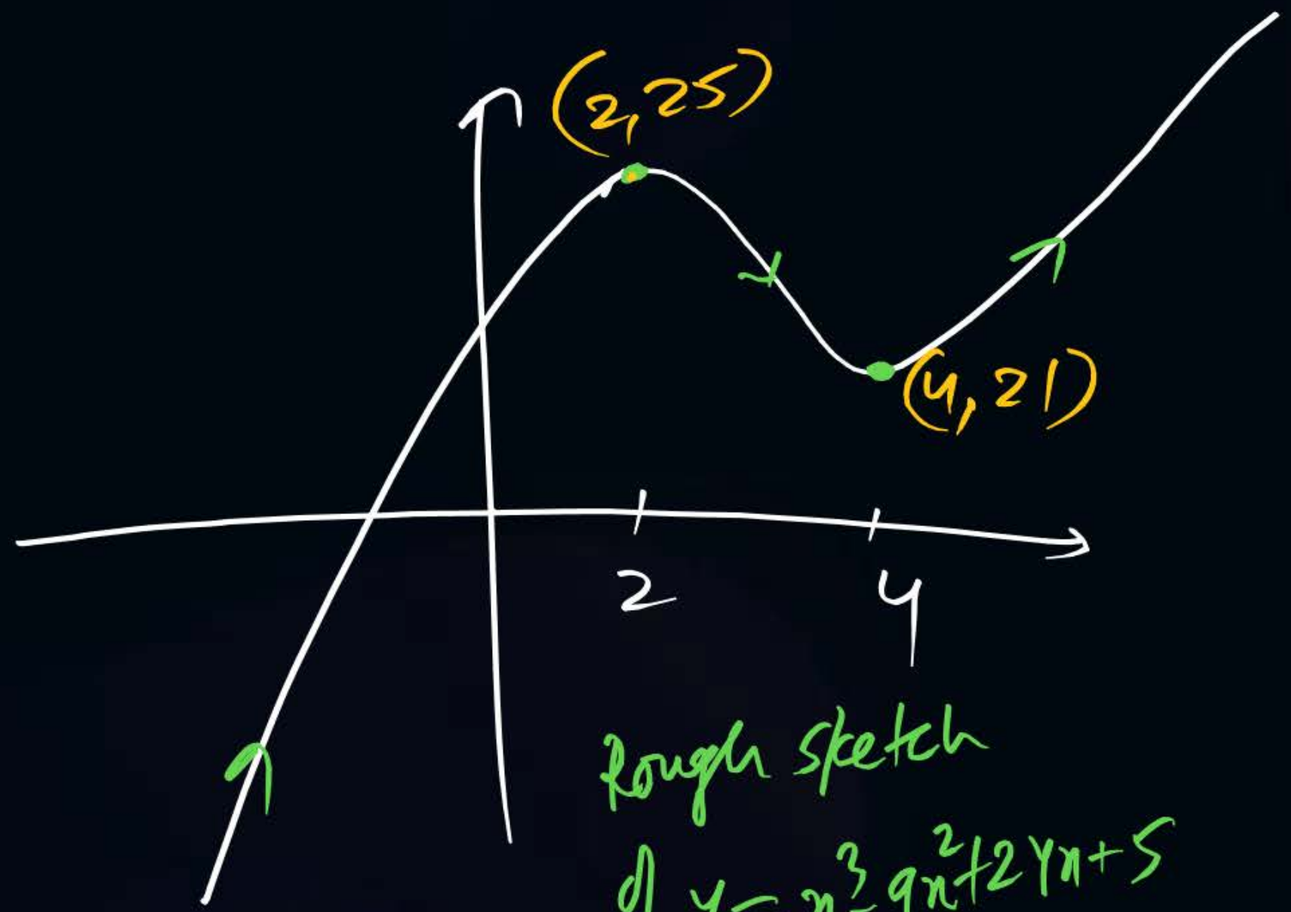
Absolute Max Value = DNE

∵ In the Right Side of $x=4$, $f(x)$ is keep on Increasing so Max Value = DNE

Local min Point is $x=4$

" " Value $y=f(4)=21$

Absolute min Value = DNE



Rough sketch

of $y = x^3 - 9x^2 + 24x + 5$

Q: The Maximum Slope of $y = x^3 - 9x^2 + 24x + 5$ will be ? = DNE

Sol: $f(x) = x^3 - 9x^2 + 24x + 5$ (given function)

$f'(x) = 3x^2 - 18x + 24$ (Slope of given func^n)

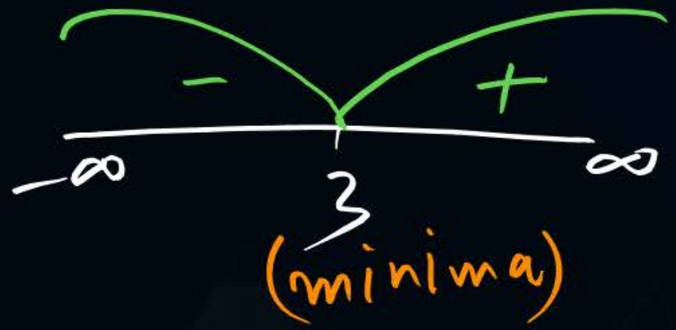
$f''(x) = 6x - 18$

Hence they are asking the Max Value of $f'(x)$ not of $f(x)$ Imp.

Now let $f'(x) = \phi(x) = 3(x^2 - 6x + 8) = 3(x-4)(x-2)$

& $f''(x) = \phi'(x) = 6(x-3)$

T. Points of $\phi(x)$ is $x = 3$



i.e Min Value of $\phi(x)$ will occur at $x = 3$ & Min Value $= \phi(3)$

$= 3(3-4)(3-2) = -3$

i.e Min slope of $f(x) = -3$

& Max ... ... = DNE

Q: $f(x)=\frac{4x^2+1}{x}$ then find the Interval in which $f(x)$ Inc & Dec. Also find minimum & Max Value of $f(x)$ if exist?

Sol:

$$f(x)=\frac{4x^2+1}{x} \quad D_f = R-\{0\}$$

$$f'(x)=\frac{x(8x)-(4x^2+1)(1)}{(x)^2}$$

$$=\frac{4x^2-1}{x^2}=\frac{(2x-1)(2x+1)}{x^2}$$

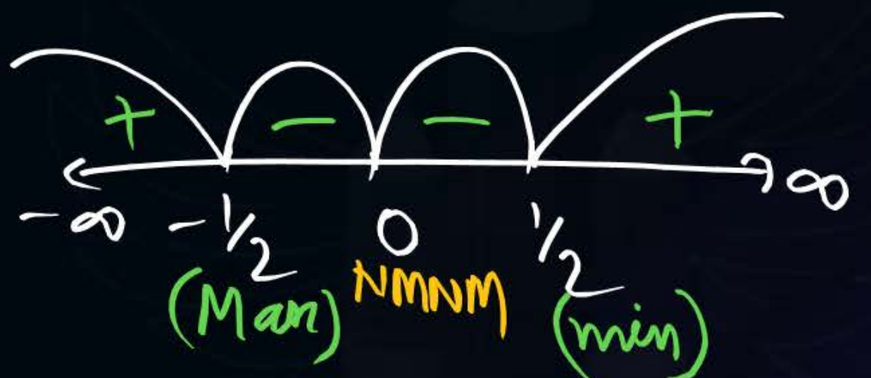
$$\boxed{f'(x)=\frac{4\left(x-\frac{1}{2}\right)\left(x+\frac{1}{2}\right)}{x^2}}$$

Critical Points are;

Either $f'(x)=0$ or $f'(x)=$ DNE

$\Downarrow$ $\Downarrow$

$x=-\frac{1}{2},\frac{1}{2}$ or $x=0$



$f(x)$ Inc in $(-\infty, -\frac{1}{2}) \cup (\frac{1}{2}, \infty)$

& $f(x)$ Dec in $(\frac{1}{2}, 0) \cup (0, \frac{1}{2})$

Local Min Point $x = 1/2$

Local Min Value $y = f(1/2) = 4$

Absolute min Value = DNE

Local Max Point $x = -1/2$

Local Max Value $y = f(-1/2) = -4$

Absolute Max Value = DNE

$$f(x) = \frac{4x^2+1}{x}$$

Don't try to Make Graphs of Longer Quest. it is Totally waste of time.

Note: $(1)^{1/3}=1$, $(-1)^{1/3}=-1$, $f(0)=(0-1)^{2/3}=((-1)^2)^{1/3}=(1)^{1/3}=1$

Q. The Minimum Value of $y=(x-1)^{2/3}$ occurs at $x=1$ & Min Value is 0

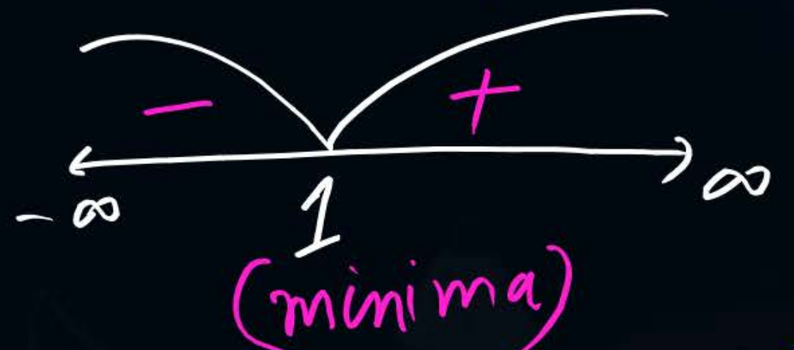
Sol: $f(x)=(x-1)^{2/3}$, $D_f=(-\infty,\infty)$

$$f'(x)=\frac{2}{3}(x-1)^{\frac{2}{3}-1}=\frac{2}{3(x-1)^{1/3}}$$

Turning Points are,

Either $f'(x)=0$ ⇓ Not able to find any Point.

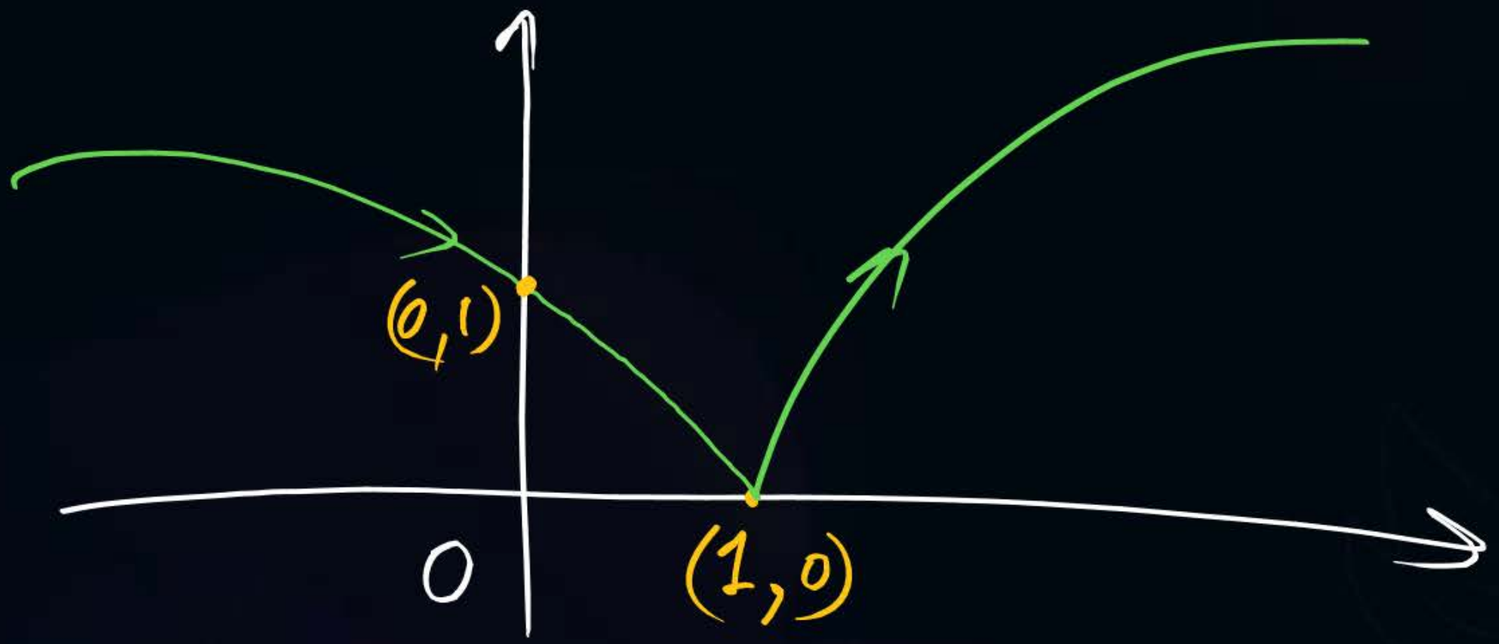
or $f'(x)=N.D.$ ⇓ $x=1$



$f(x)$ will be minimum at $x=1$ & Minimum Value is $y=f(1)=$

$=(1-1)^{2/3}=0$

Graph: $y=(x-1)^{2/3}$, $D_f=(-\infty,\infty)$



$f(x)=(x-1)^{2/3}$

$f'(x)=\frac{2}{3(x-1)^{1/3}}$

$f''(x)=\frac{-2}{9(x-1)^{4/3}}$

$\because f''(x)<0$ always

$\Rightarrow f(x)$ is Concave Down.

Q: For the func $e^y = x^{1/x}$, Max Value occurs at ? $x = e$
& Max Value will be ? $y = \frac{1}{e}$

Sol: $\log_e(e^y) = \log(x)^{1/x}$

$y = \frac{1}{x}\log x$

$\boxed{f(x) = \frac{\log x}{x}}$, $D_f : (0, \infty)$

Already Asked Yesterday.

T. Point is $x = e$

Point of Maxima is $x = e$

& Max Value $= f(e) = \frac{\log e}{e} = \frac{1}{e}$

Q: $f(x)=\dfrac{e^{\sin x}}{e^{\cos x}}$, $x\in R$ then Max Value of $f(x)$ is _____

Sol: $f(x)=e^{\sin x-\cos x}$

$f'(x)=e^{\sin x-\cos x}(\cos x+\sin x)$

Turning points are, $f'(x)=0$

$e^{(\sin x-\cos x)}(\cos x+\sin x)=0$

$\because e^{\sin x-\cos x}\neq 0$ so $\cos x+\sin x=0$

$\sin x=-\cos x$

$\tan x=-1$

$\tan x=-1$

$\Rightarrow x=\ldots -\frac{\pi}{4}, \frac{3\pi}{4}\ldots\ldots$



i.e Max Value occurs at $x=\frac{3\pi}{4}$

& Max Value $=f\left(\frac{3\pi}{4}\right)=e^{\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}}=e^{\sqrt{2}}$

Q8 (Type II) A Right Circular Cone of Maximum Volume is to be inscribed in a sphere of Radius 1 mtr then find the Height of such Cone.
(Application Based Questions)

Ans: $h = \frac{4}{3}$

THANK - YOU

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